

Supply Chain Visibility & Optimisation

Complete Detailed Project Documentation (Milestones 1 - 4)

Project Type	Team Project
Team Number	4
Team Members	Mahi, Ramya, Vishal, Abdu Haseeb
Mentor Name	Sundar Shakthi
Internship / Program	Infosys Springboard Virtual Internship 7.0 - Batch 2
Date	17 August 2026
Prepared by	Abdul Haseeb

Table of Contents

1. Executive Summary & Project Context
2. Milestone 1: Data Preprocessing & Validation
3. Milestone 2: Power BI Control Tower & Semantic Model
4. Milestone 3: Performance Analytics & Decision Support (Overview)
5. Milestone 3 Workstream 1: Supplier Performance Scorecards
6. Milestone 3 Workstream 2: Transportation Cost Analysis
7. Milestone 3 Workstream 3: Route & Carrier Performance
8. Milestone 3 Workstream 4: Advanced KPI Visualization
9. Milestone 4: Integrated Business Insights
10. Milestone 4: Strategic Recommendations & Action Plan
11. Analytical Limitations & Caveats
12. Conclusion & Final Sign-Off

1. Executive Summary & Project Context

The modern e-commerce supply chain generates massive volumes of transactional data, spanning orders, payments, shipping, and customer reviews. Without a centralized analytical layer, this data remains fragmented and unable to support operational decision-making. The core business objective of this project was to transform raw transactional e-commerce data into a practical, decision-ready supply-chain control tower.

By unifying product movement, delivery reliability, freight expenditure, and seller performance into a single Power BI environment, the project enables management to identify systemic bottlenecks, evaluate vendor reliability, and optimize network transportation costs. This documentation traces that entire journey from raw Python data cleaning through to advanced cross-KPI management views.

2. Milestone 1: Data Preprocessing & Validation

2.1 Raw Dataset Profiling

The project originated with the Olist Brazilian E-Commerce dataset. The raw, merged dataset contained exactly 119,143 records spanning 40 distinct columns. The initial data profiling phase utilizing Python (Pandas) revealed significant noise, including missing values in critical dimensions (such as product categories and delivery timestamps) and extreme outliers that would severely distort average-based KPIs.

2.2 Data Cleaning Workflow

A rigorous six-step Python workflow was executed to ensure absolute data integrity prior to visualization:

- 1. Profile: Analysed matrix shape, data types, null counts, and exact duplicate rows (0 exact duplicate rows found).
- 2. Clean: Dropped unrecoverable noise and imputed missing categorical/numeric gaps logically.
- 3. Validate: Assessed price distributions, freight consistency, and physical product dimensions.
- 4. Engineer: Created net-new operational features to support delivery analytics.
- 5. Treat Outliers: Applied strict statistical boundaries to isolate extreme exceptions.
- 6. Export: Finalized the 'Supply_Chain_Cleaned_Dataset.csv' for Power BI ingestion.

2.3 Feature Engineering: Delivery Delay

A critical requirement for measuring supply chain reliability was comparing actual delivery performance against the customer's expectation. A new feature was engineered:

```
delivery_delay = delivered_customer_date - estimated_delivery_date
```

Negative values indicate an early delivery (exceeding customer expectation), while positive values represent late deliveries. This feature became the foundation for the On-Time Delivery KPIs.

2.4 Outlier Treatment (IQR Methodology)

The dataset contained extreme values that could skew average metrics. The Interquartile Range (IQR) method was applied systematically across numeric distributions:

```
IQR = Q3 - Q1
Lower Bound = Q1 - (1.5 * IQR)
Upper Bound = Q3 + (1.5 * IQR)
```

Observations falling outside these bounds were filtered to ensure the core dataset represented standard operating conditions rather than extreme anomalies.

2.5 Cleaned Output & Analytical Grain

Following preprocessing, the dataset was reduced from 119,143 raw rows to 81,170 pre-outlier rows, and ultimately to 66,039 fully cleaned rows across 36 final columns. This validated output represents 55,649 distinct customer orders and 63,089 unique order-item movement lines. The analytical grain established here remained the strict foundation for every KPI developed in the subsequent milestones.

3. Milestone 2: Power BI Control Tower & Semantic Model

3.1 Duplicate-Safe Logic & Data Grain

During semantic modelling, a critical risk was identified: the merged dataset repeats order-item rows when an order possesses multiple payment records (e.g., paying with a combination of credit card and voucher). A naive SUM(price) would artificially inflate total sales by counting the same item multiple times.

To resolve this, the team defined a unique Movement Line Key (order_id + order_item_id). The duplicate-safe DAX logic evaluates sales exclusively at this unique grain:

```
Total Sales =
SUMX (
    VALUES ('Supply_Chain_Cleaned_Dataset'[Movement Line Key]),
    CALCULATE (MAX ('Supply_Chain_Cleaned_Dataset'[price]))
)
```

Similarly, physical movement was tracked using a distinct count to avoid duplication:

```
Units Moved =
DISTINCTCOUNT ('Supply_Chain_Cleaned_Dataset'[Movement Line Key])
```

3.2 Inventory-Proxy Methodology

Mandatory methodology statement: Product movements are used as an inventory proxy; the dataset does not contain actual warehouse stock or on-hand inventory balances.

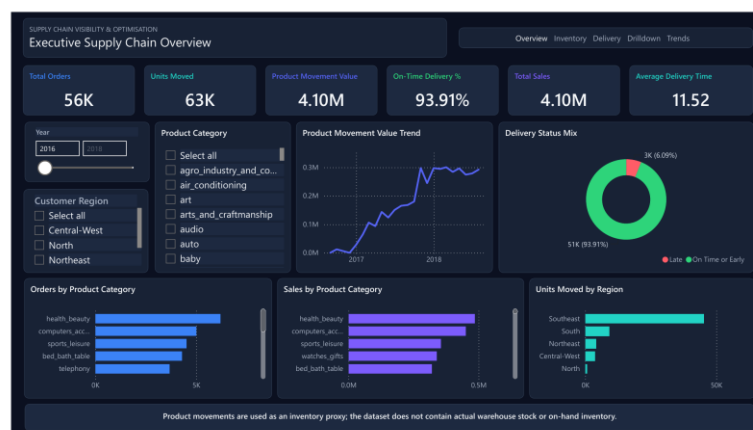
Because the public Olist dataset lacks warehouse stock balances, stock snapshots, or reorder points, the project avoids claiming actual inventory availability. Instead, all inventory-oriented analysis is based on observed order-item movement volume. Metrics such as Product Velocity, Active Products, and Demand Variability serve as highly effective proxies for inventory turnover and product lifecycle analysis.

An ABC Classification system was implemented based on cumulative Product Movement Value contribution. Class A covers products contributing up to 80% of cumulative value, separating high-value product movement from lower-value long-tail movement (Classes B and C).

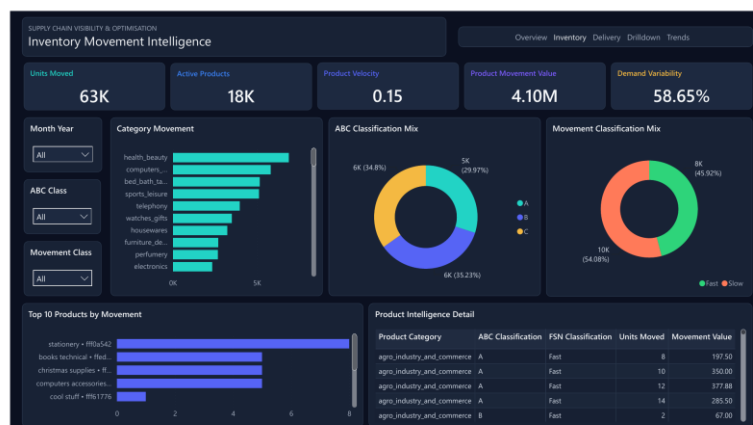
3.3 The Five Visibility Dashboards

The cleaned data was visualised across five distinct control tower pages:

- 1. Executive Supply Chain Overview: A flagship page tracking Total Orders (56K), Units Moved (63K), Product Movement Value (4.10M), and On-Time Delivery (93.91%).



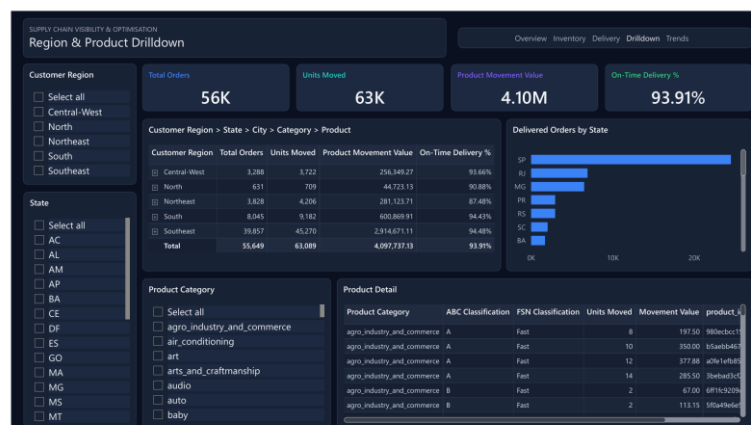
- 2. Inventory Movement Intelligence: Tracks ABC Classification, Fast/Slow moving category movement, and Demand Variability (58.65%).



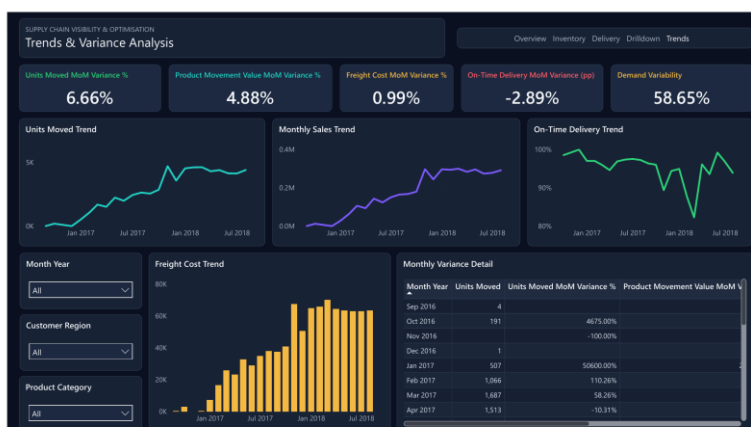
- 3. Delivery Performance: Evaluates operational reliability, highlighting Average Delivery Time (11.52 days), Average Late Delivery Days (10.14 days), and comparing On-Time versus Late delivery clusters.



- 4. Region & Product Drilldown: Provides an investigative hierarchy allowing users to drill from Customer Region -> State -> City -> Product Category -> Individual Product.



- 5. Trends & Variance Analysis: Monitors Month-over-Month (MoM) directional changes in movement, sales, and freight, using rigorous Date-table logic to compute chronological variances securely.



4. Milestone 3: Performance Analytics & Decision Support (Overview)

Milestone 3 fundamentally transitions the project from answering 'What happened?' (Descriptive Visibility) to asking 'Where is performance strong, weak, costly, or inefficient?' (Diagnostic & Performance Analytics).

The objective was to leverage the validated Power BI model to answer practical management queries:

- Which sellers are performing well and which require immediate operational attention?
- Where is transportation money being spent, and does higher spend yield better delivery times?
- Which geographic routes carry the highest movement volume?
- How does straight-line distance affect transit performance expectations?
- How can freight, delivery, customer reviews, and other KPIs be combined into cohesive management views?

This was accomplished via four distinct analytical workstreams: Supplier Performance Scorecards, Transportation Cost Analysis, Route & Carrier Performance, and Advanced KPI Visualization.

5. Milestone 3 Workstream 1: Supplier Performance Scorecards

5.1 Objective and Eligibility

The Supplier Performance Scorecard was created to evaluate individual sellers holistically across multiple dimensions—Delivery reliability, Customer review/experience, Freight efficiency, and Delivery speed—rather than judging them on a single, potentially misleading metric.

To ensure statistical fairness and prevent sellers with negligible volume from distorting the rankings, a strict minimum-volume eligibility rule was applied: Only sellers with at least 50 delivered orders were included in the formal ranking. This isolated 235 eligible sellers for deep-dive analysis.

5.2 Scoring Logic & DAX

The final overall score combines the four component scores equally, preventing any single dimension (like a high customer review) from masking a critical flaw (like abysmal freight efficiency).

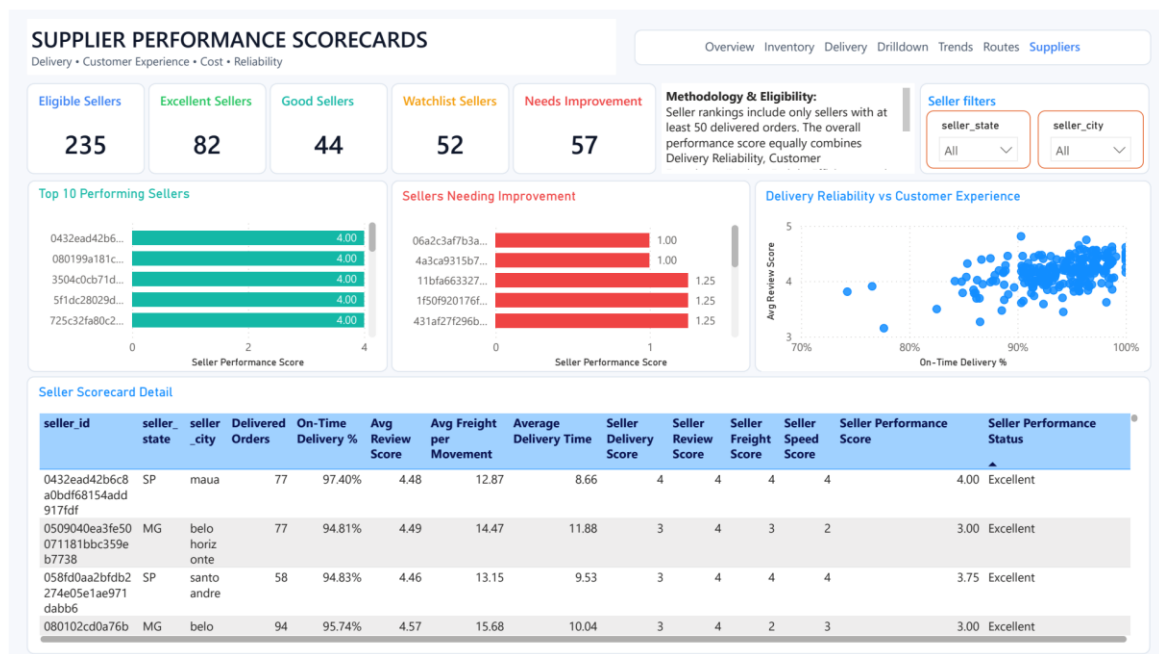
```
Seller Performance Score =  
DIVIDE(  
    [Seller Delivery Score]  
    + [Seller Review Score]  
    + [Seller Freight Score]  
    + [Seller Speed Score],
```

5.3 Performance Bands & Findings

Based on this robust scoring logic, the 235 eligible sellers were classified into quartile-based performance bands:

- Excellent (Score ≥ 3.00): 82 sellers
- Good (Score ≥ 2.50): 44 sellers
- Watchlist (Score ≥ 2.00): 52 sellers
- Needs Improvement (Score < 2.00): 57 sellers

Key Finding: 109 sellers (52 Watchlist + 57 Needs Improvement) fall into the bottom tiers. This represents approximately 46.4% of the eligible seller population. One of the strongest conclusions of this project is that almost half of the active, high-volume seller base requires dedicated performance governance.



6. Milestone 3 Workstream 2: Transportation Cost Analysis

6.1 Objective & Main KPI

The Transportation Cost Analysis dashboard shifts focus to financial expenditure, identifying where transportation capital is being spent and examining its correlation with delivery performance. The dashboard calculates a Total Transportation Cost of approximately 969.23K.

```
Total Transportation Cost =
SUM(freight_value) // Extended to Movement Line Key grain to prevent
duplicates
```

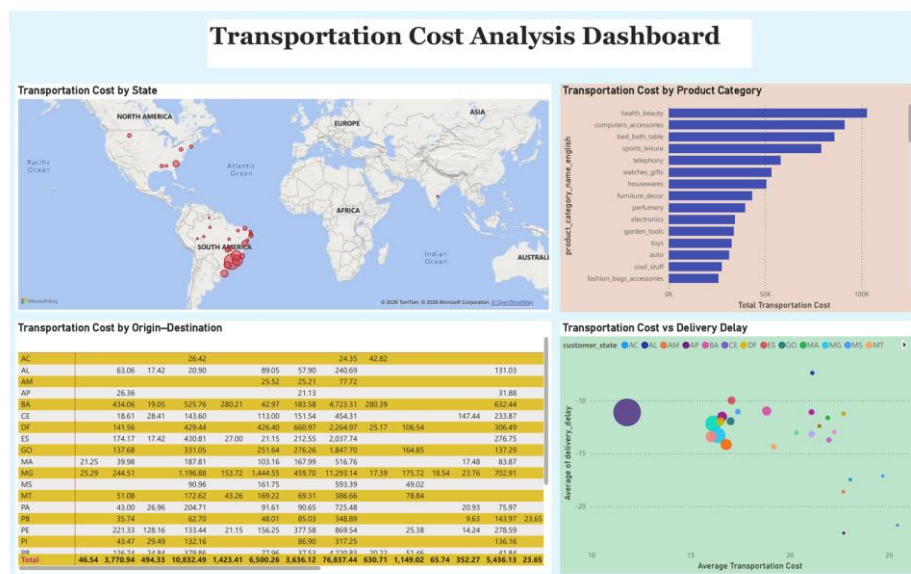
6.2 Geographic & Categorical Concentration

Transportation spend is highly asymmetrical. The geographic analysis pinpoints specific states as disproportionately large cost drivers, notably RJ (approx. 137.94K) and MG (approx. 127.44K). Furthermore, analysis by product category reveals that 'Health & Beauty' and 'Computers & Accessories' dominate the freight expenditure.

Crucially, the dashboard provides an origin-destination matrix. This allows management to move past vague questions like 'Which state is expensive?' to actionable queries like 'Which specific lane between origin state X and destination state Y is inflating our freight costs?'

6.3 Cost vs. Inefficiency

A dedicated scatter plot maps Average Transportation Cost against Average Delivery Delay. This visual reinforces a critical business reality: Cost $\neq$ Inefficiency by itself. A high-cost route may be perfectly justified if it yields exceptionally fast and reliable delivery. By plotting these variables together, management can isolate quadrants where high cost intersects with high delay—the true zones of operational inefficiency.



7. Milestone 3 Workstream 3: Route & Carrier Performance

7.1 Methodological Context

The Route & Carrier Performance dashboard evaluates logistics movement at the route level (Seller State -> Customer State). Due to a limitation in the Olist dataset—the absence of a reliable carrier identity field—'carrier performance' is defined as the transit time between

Carrier Handoff and Customer Delivery. Furthermore, route distance is calculated as an approximate straight-line geographic distance using Haversine coordinates (Earth radius ~6,371 km), not exact road routing.

7.2 Route Volumes & Distance Effects

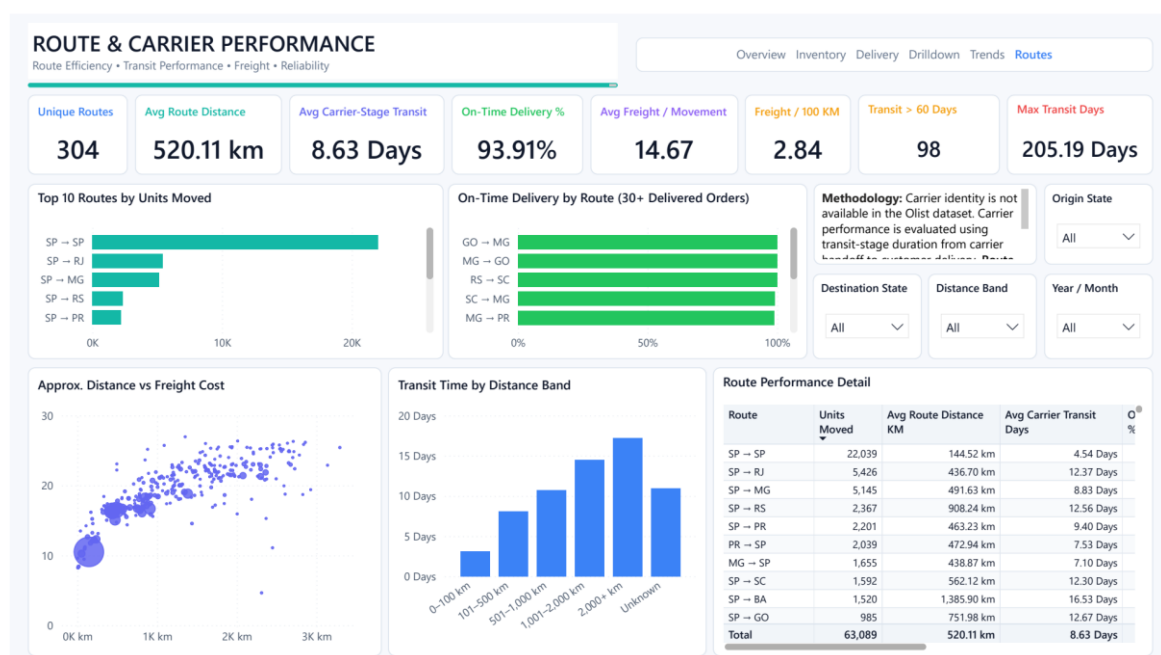
The dashboard tracks 304 unique routes with an average route distance of 520.11 km and an average carrier-stage transit of 8.63 days. Network movement is heavily dominated by São Paulo; the SP -> SP lane alone carries 22,039 units, followed by SP -> RJ (5,426) and SP -> MG (5,145).

Distance strongly influences transit times. For instance, the SP -> SP route (~144 km) averages 4.54 days, while the much longer SP -> BA route (~1,385 km) averages 16.53 days. This evidence proves that applying a single, global delivery SLA target across the entire network is operationally unfair and misleading.

7.3 Exception Monitoring & DAX

While average on-time delivery is high (93.91%), averages can mask extreme failures. The dashboard flags 98 specific orders where carrier-stage transit exceeded 60 days, peaking at a maximum transit of 205.19 days. These exceptions require a dedicated, separate management workflow outside of standard KPI tracking.

```
Total Freight Cost = SUMX (VALUES ([Movement Line Key]),  
MAX ([freight_value]))  
Freight Cost per 100 KM = DIVIDE ([Total Freight Cost], [Total Route  
Distance KM]) * 100  
Avg Carrier Transit Days = AVERAGEX (VALUES ([order_id]), MAX ([Carrier  
Transit Days]))
```



8. Milestone 3 Workstream 4: Advanced KPI Visualization

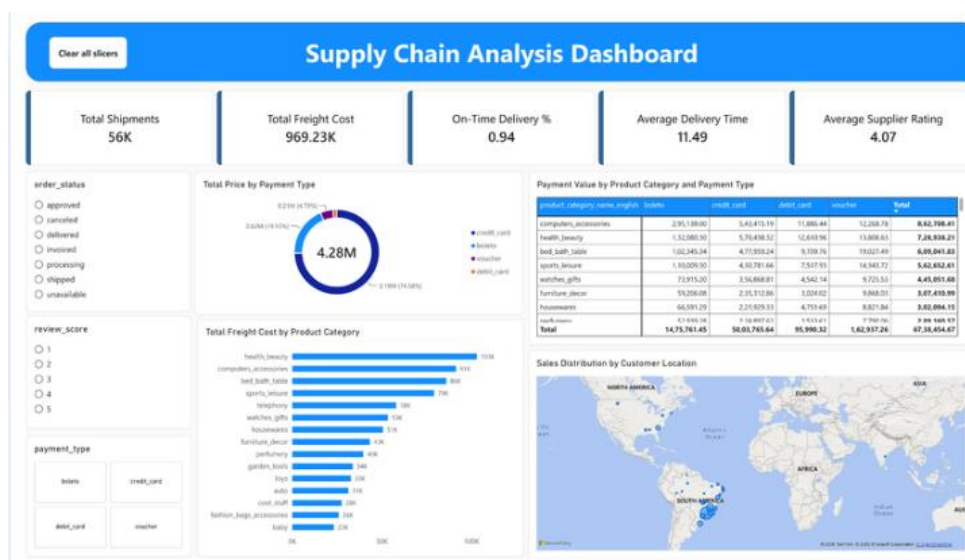
8.1 Management Overview

The Advanced KPI dashboards serve as the ultimate executive summary, converging shipment volume, freight, delivery speed, supplier ratings, payment mixes, and geography into a single pane of glass. It is designed for quick management consumption before deep-diving into the diagnostic workstreams.

Core Executive KPIs:

- Total Shipments: 56K
- Total Freight Cost: 969.23K
- On-Time Delivery: 0.94 (~94%)
- Average Delivery Time: 11.49 Days
- Average Supplier Rating: 4.07

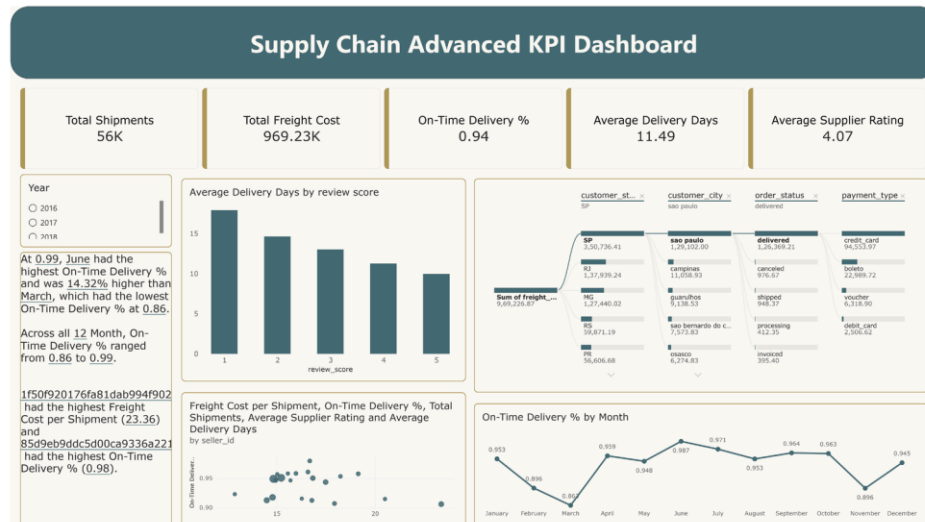
```
On-Time Delivery % =  
DIVIDE (  
    CALCULATE (  
        DISTINCTCOUNT('Supply_Chain_Cleaned_Dataset'[order_id]),  
        'Supply_Chain_Cleaned_Dataset'[delivery_delay] <= 0  
    ),  
    [Total Shipments]  
)
```



8.2 Service Pattern Analysis

The second view within this workstream identifies longitudinal service patterns. While the aggregate on-time delivery is ~94%, monthly reliability fluctuates between 86% and 99%. The narrative insight explicitly identifies March as the weakest service month and June as the strongest, providing clear targets for seasonality and root-cause analysis.

Additionally, a correlation bar chart maps Average Delivery Days against Review Scores. A clear, descriptive relationship emerges: average delivery days decline consistently as customer review scores increase. This quantifies the direct impact of supply chain efficiency on end-customer satisfaction.



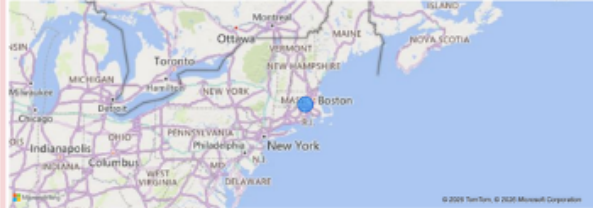
9. Milestone 4: Integrated Business Insights

Milestone 4 represents the final synthesis of the project. By evaluating the descriptive control tower (M2) and the diagnostic workstreams (M3) collectively, six critical business insights are formalized:

- 1. Service is strong overall, but averages hide flaws: The 93.91% on-time delivery rate is a solid baseline, but local bottlenecks and extreme exceptions exist beneath the surface.
- 2. Seller improvement is a massive opportunity: With 46.4% of eligible sellers requiring monitoring (Watchlist/Needs Improvement), aggressive vendor governance will yield immediate network-wide benefits.
- 3. Cost is geographically concentrated: Transportation spend is highly uneven. High-cost states (RJ, MG) and specific lanes must be scrutinized through a cost-to-service lens.
- 4. Distance fundamentally shapes transit behavior: Transit time scales with distance bands. Global delivery targets are inadequate; regional, distance-adjusted SLAs are required.
- 5. Exceptions need dedicated workflows: Averages are distorted by 205-day extreme delays. A separate exception-management process is mandatory.
- 6. Customer experience is tethered to operations: Shorter delivery durations strictly align with higher customer review scores, proving that logistics speed drives brand perception.

Executive Overview

Order Status Distribution



37.59K

Total Order Value

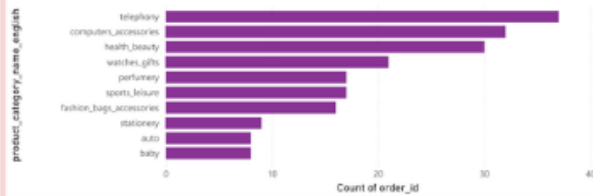
223

Total Orders

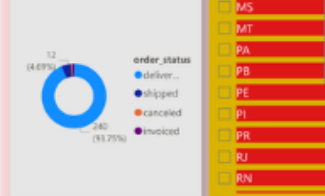
customer_state



Top 10 Product Categories by Orders



Orders by Customer State



Delivery Efficiency Dashboard

Total Shipments

56K

On-Time Delivery %

94%

Average Delivery Days

11.49

Average Supplier Rating

4.07

Freight Cost per Shipment

17.42

order_status

- ☐ approved
- ☐ cancelled
- ☐ delivered
- ☐ invoiced
- ☐ processing
- ☐ shipped
- ☐ unavailable

Year

2016 2017 2018

payment_type

- ☐ bank
- ☐ credit_card
- ☐ debit_card
- ☐ voucher

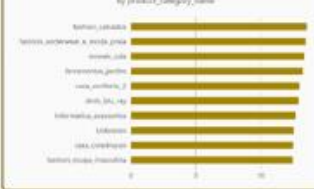
Average Delivery Time by Review Score



Monthly On-Time Delivery Efficiency



Average Delivery Days by product_category_name

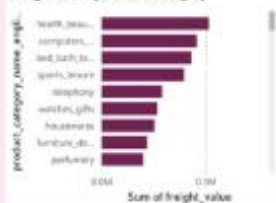


Total Shipments



Transportation Cost Performance Overview

Freight Cost by Product Category



Product Price vs. Freight Cost



14.68

Average Freight Cost

969.23K

Total Freight Cost

64.81

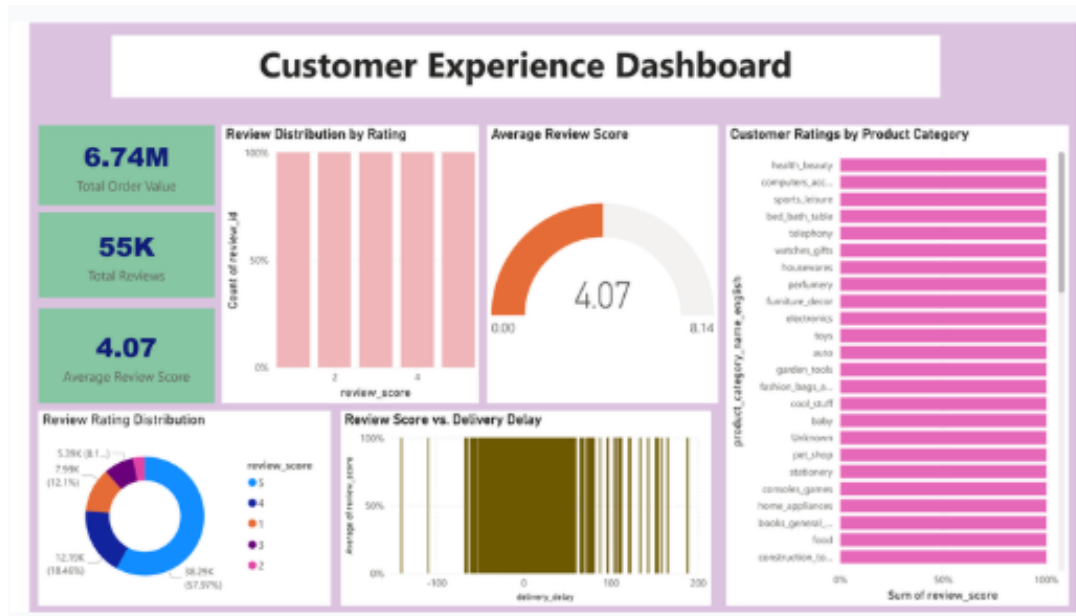
Average Product Price

Freight Cost by Customer State



Freight Cost by Seller State





10. Milestone 4: Strategic Recommendations & Action Plan

To translate these insights into a focused operating rhythm, the following strategic actions are recommended to management:

01. Implement Supplier Governance (Priority: HIGH)

Establish a monthly scorecard review rhythm specifically targeting the 109 sellers in the Watchlist and Needs Improvement quartiles. Rather than general feedback, address their weakest specific component (e.g., freight efficiency vs. delivery speed) using the scorecard metrics.

02. Enforce Route Exception Monitoring (Priority: HIGH)

Configure automated alerts to flag carrier-stage transit times exceeding 60 days. Divert these exceptions from standard KPI dashboards into a dedicated root-cause investigation workflow to resolve dead-in-transit orders.

03. Execute Freight Optimisation (Priority: HIGH)

Conduct a deep-dive investigation into high freight-per-distance lanes (e.g., specific RJ or MG routes) and categories (Health & Beauty). Renegotiate terms or find alternative logistics partners where high spend is failing to produce corresponding delivery speed improvements.

04. Establish Service Benchmarks by Distance (Priority: MEDIUM)

Abandon the single network-wide transit target. Establish dynamic, tiered transit expectations based on the distance bands visualized in the dashboard (0-100 km, 101-500 km, 501-1,000 km, etc.) to evaluate carriers fairly.

05. Standardize KPI Operating Cadence (Priority: MEDIUM)

Adopt the Advanced KPI overview page as the standard opening view for all monthly management meetings. Use it to quickly connect volume, freight, delivery, and reviews before drilling down into problematic routes or suppliers.

11. Analytical Limitations & Caveats

To maintain analytical integrity and trust, the following data limitations are explicitly acknowledged:

- No actual warehouse inventory: Because the dataset lacks stock-on-hand or reorder points, all inventory metrics utilize observed product movement as a proxy.
- No named carrier identity: The dataset lacks carrier vendor names. 'Carrier performance' strictly measures the transit phase from seller handoff to customer delivery.
- Approximate distance logic: Route distances are derived from Haversine straight-line coordinates between state centroids, not exact road-network topography.
- Historical context: This is a diagnostic and descriptive analysis of historical data; it does not utilize machine learning to forecast future demand or predict specific delivery failures.
- Anonymized products: Commercial product names were anonymized in the source data; analysis relies heavily on categorical roll-ups and hashed product IDs.

12. Conclusion & Final Sign-Off

The Supply Chain Visibility & Optimisation project has successfully executed its mandate. Beginning with 119,143 raw, disjointed e-commerce records, Milestone 1 established a pristine, 66,039-row foundation. Milestone 2 constructed a robust Power BI semantic model featuring duplicate-safe logic, materializing in five descriptive visibility dashboards.

Milestone 3 elevated the project from visibility to diagnostic intelligence, deploying supplier scorecards, transportation cost matrices, route performance analytics, and advanced KPI views. Finally, Milestone 4 synthesized these findings into actionable, performance-focused decision support.

By unifying Cost + Service + Seller Performance + Route Reliability, this control tower moves beyond merely reporting historical numbers. It empowers management to identify exactly where the network is strong, where capital is being wasted, and which specific operational levers must be pulled to drive supply chain excellence.